Longevity orthologs across four major model organisms

Complete visual catalogue • 320 genes • 206 direct pairs

Reviewed example = a checked experiment; GenAge annotation = catalogue evidence only.

All genes in the qualifying pair set are included; rows organise direct links, not proven conserved mechanisms.

	Budding yeast	Nematode	Fruit fly	Mouse
C114 **GSY2** 2 genes / 2 species 1 direct pair	**Y1 GSY2** GenAge 1364 **GenAge annotation** **Necessary for fitness** Lifespan: Decrease Refs: 20883526	No gene in this candidate network	**F1 GlyS** GenAge 1919 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 25059425	No gene in this candidate network
Direct pairs: F1–Y1 (1:m)				
C128 **GTR1** 2 genes / 2 species 1 direct pair	**Y1 GTR1** GenAge 1746 **GenAge annotation** **Necessary for fitness** Lifespan: Decrease Refs: 22912585	**W1 raga-1** GenAge 2062 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 22560223	No gene in this candidate network	No gene in this candidate network
Direct pairs: W1–Y1 (1:1)				
C039 **hsp-4** 2 genes / 2 species 1 direct pair	No gene in this candidate network	**W1 hsp-4** GenAge 345 **GenAge annotation** **Pro-Longevity** Lifespan: Decrease Refs: 16741121	**F1 Hsc70-3** GenAge 853 **GenAge annotation** **Pro-Longevity** Lifespan: Increase Refs: 18059160	No gene in this candidate network
Direct pairs: W1–F1 (m:1)				
C112 **HXT5** 2 genes / 2 species 1 direct pair	**Y1 HXT5** GenAge 1239 **GenAge annotation** **Necessary for fitness** Lifespan: Decrease Refs: 20421943	**W1 fgt-1** GenAge 1912 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 24024580	No gene in this candidate network	No gene in this candidate network
Direct pairs: W1–Y1 (n:m)				
C028 **IDH2** 2 genes / 2 species 1 direct pair	**Y1 IDH2** GenAge 358 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 16293764, 18340043 19030232	**W1 idha-1** GenAge 260 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 15998808	No gene in this candidate network	No gene in this candidate network
Direct pairs: W1–Y1 (1:1)				

Y/W/F/M identify genes in yeast/worm/fly/mouse columns. Pair codes retain the OMA source orientation.
Refs are catalogue-linked PMIDs, not newly reviewed experiments. Empty cells do not establish biological absence.
GenAge lifespan labels do not specify interventions. Only reviewed examples have experiment arrows and colours.
Sources: GenAge Build 21; OMA All.May2026. See atlas legend, gene index and source tables for evidence details.